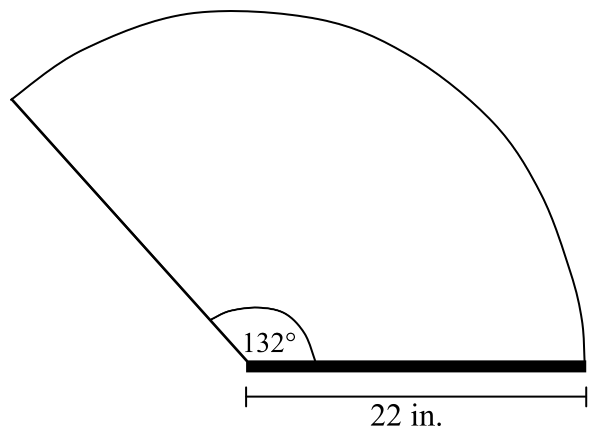
Question 1

2 marks 101

A section of a car windshield is cleaned by a wiper as shown in the diagram below. The arm of the wiper is 22 inches, and it rotates through a central angle of $132°$. Determine the length of the arc that is created by the tip of the wiper.



Question 2 — 2 marks 102

There are 20 boys and 11 girls who can be selected to be on a team.

Determine the number of ways that 7 boys and 5 girls can be selected for this team.

Question 3 — 2 marks 103

A water filtration system which removes impurities from a sample of water can be modelled by $P = 0.25(0.55)^n$, where:

$P =$ the percentage of impurities remaining, in decimal form

$n =$ the number of filters

Determine, algebraically, how many filters are required so that less than 1% of the impurities remain in the water sample. Express your answer as a whole number.

Question 4 — 3 marks

In the binomial expansion of $\left(x^2 - \frac{2}{y}\right)^8$, determine the middle term in simplified form.

Question 5 3 marks 105

Solve the following equation algebraically over the interval $[0, 2\pi]$.

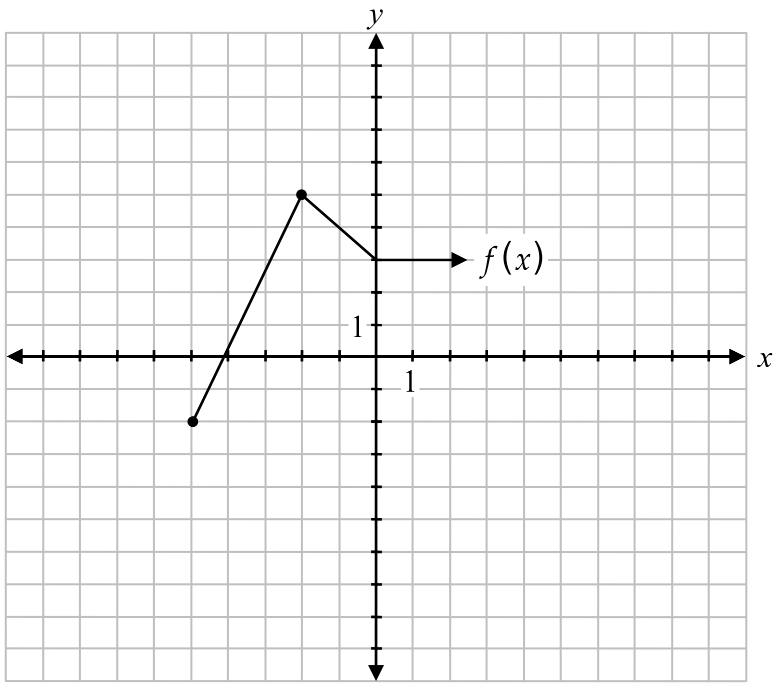
$$6\sin^2\theta + \sin\theta - 1 = 0$$

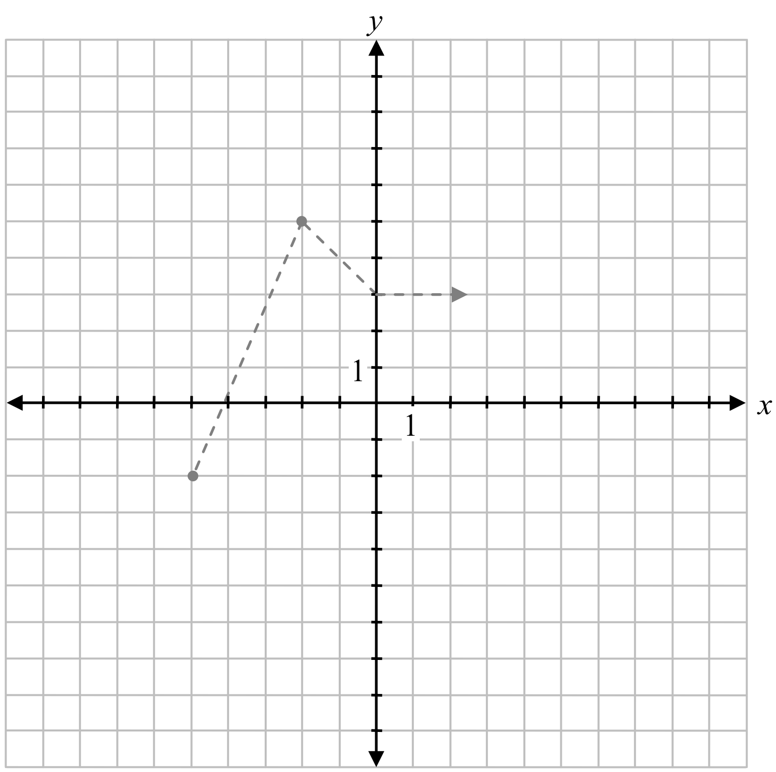
Note: A calculator is not required for the remaining test questions.

Question 6 2 marks

106

Given the graph of $y = f(x)$, sketch the graph of $y = f(-x+4)$.



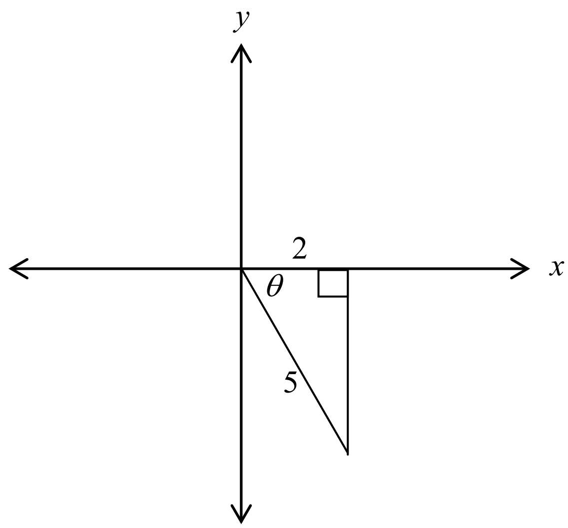


The graph of $f(x)$ has already been drawn for your reference.

No marks will be awarded for the graph of $f(x)$.

Question 7 2 marks

Given the following triangle, determine $\csc\theta$.



Question 8 3 marks 108

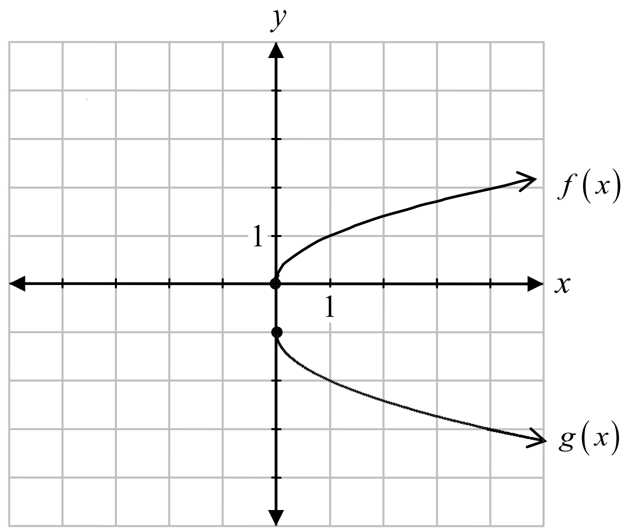
Solve algebraically.

$$_nP_2 = 9n$$

Question 9 2 marks

109

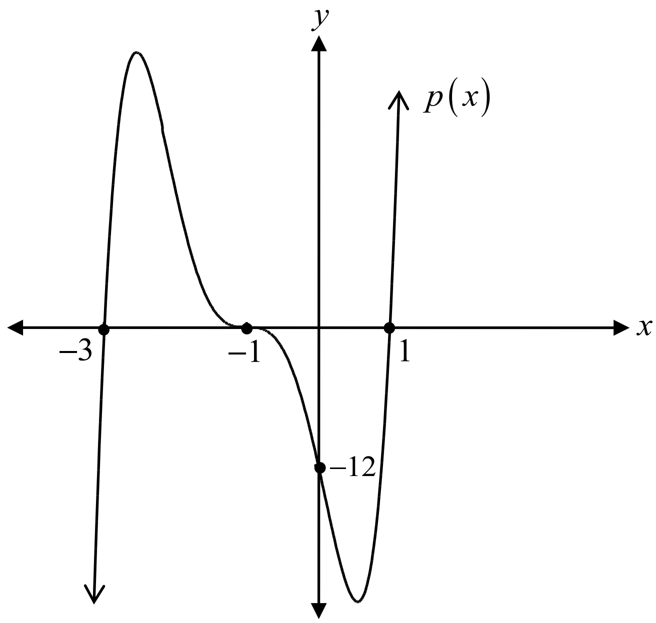
Describe the transformations applied to the graph of $f(x)$ to obtain the graph of $g(x)$.



Question 10

2 marks 110

Determine, algebraically, the value of the leading coefficient of the graph of the polynomial function, $p(x)$.



Question 11 1 mark 111

Frank, Liam, Chan, and Thao are going to a movie.

Determine the number of ways they can sit in a row of four chairs, if Frank and Chan must sit beside each other.

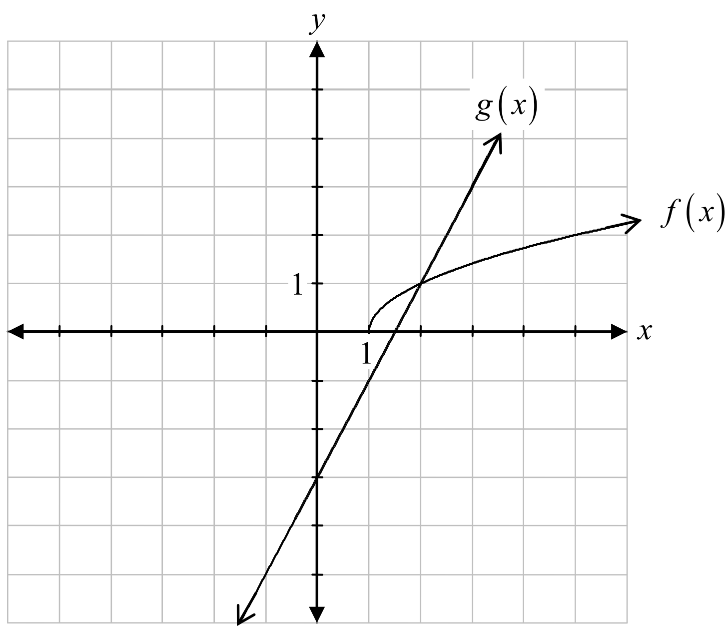
Question 12 2 marks 112

Determine, algebraically, if $f\left(x\right)=\dfrac{1}{x+5}$ and $g\left(x\right)=\dfrac{1}{x-5}$ are inverses of each other.

Justify your answer.

Question 13 1 mark

Using the graphs of $y = f(x)$ and $y = g(x)$, solve $f(x) = g(x)$.



Question 14 1 mark

114

An angle in standard position measures $\frac{3\pi}{4}$.

Determine in which quadrant the terminal arm of this angle is located after a rotation of 3 radians.

Justify your answer.

Question 15 3 marks 115

Prove the following identity for all permissible values of θ.

$$\frac{\sin 2\theta}{1-\cos 2\theta} = \cot \theta$$

Left-Hand Side	Right-Hand Side

Question 16 1 mark 116

If the range of $y = f(x)$ is $-3 \le y \le 6,$ determine the range of $y = 2f(3x)$.

Question 17 1 mark 117

Maurice incorrectly solved the equation, $\sin\theta + 1 = 0,$ over the interval $[0^\circ, 360^\circ]$.

$$\sin\theta + 1 = 0$$
$$\sin\theta = -1$$
$$\sin\theta = 270^\circ$$

Describe his error.

No marks will be awarded for work done on this page.

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